Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2019 – Paper 2 – Solutions**

June 2019 – Question 1 – Solution

(a) The quadratic expression $f(x) = ax^2 + 12x + b$ is divisible by $x - 3$ and has a remainder of -27 when divided by $x + 6$.

(i) Show that the values of the constants a and b are 3 and -63, respectively. [7]

$f(x) = ax^2 + 12x + b$

Since $(x - 3)$ is a factor of $f(x)$, then by the factor theorem, $f(3) = 0$.

Well,

$f(3) = a(3)^2 + 12(3) + b$

$0 = 9a + 36 + b$

$b = -9a - 36 \qquad \rightarrow$ Equation 1

Since $f(x)$ leaves a remainder of -27 when divided by $(x + 6)$, then by the remainder theorem, $f(-6) = -27$.

Well,

$f(-6) = a(-6)^2 + 12(-6) + b$

$-27 = 36a - 72 + b$

$b = -36a + 45 \qquad \rightarrow$ Equation 2

Equating Equation 1 and Equation 2 gives:

$-9a - 36 = -36a + 45$

$36a - 9a = 45 + 36$

$27a = 81$

$a = 3$

Substituting $a = 3$ into Equation 1 gives:

$b = -9(3) - 36$

$= -27 - 36$

$= -63$

$\therefore a = 3$ and $b = -63$

Q.E.D.

(ii) Hence, determine the factors of f. [3]

$f(x) = 3x^2 + 12x - 63$

When $f(x) = 0$,

$3x^2 + 12x - 63 = 0$

$(\div 3)$

$x^2 + 4x - 21 = 0$

$(x + 7)(x - 3) = 0$

$\therefore$ The factors are $(x + 7)$ and $(x - 3)$.

(b) Solve, for real values of x, the inequality $|3x - 4| \leq 6$. [4]

$|3x - 4| \leq 6$

By definition, either $3x - 4 \leq 6$ or $-(3x - 4) \leq 6$.

<u>Case 1:</u>

$3x - 4 \leq 6$

$3x \leq 10$

$x \leq \frac{10}{3}$

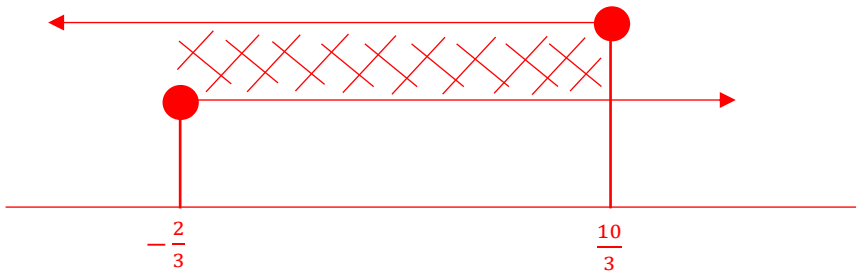
<u>Case 2:</u>

$-(3x - 4) \leq 6$

$-3x + 4 \leq 6$

$-3x \leq 2$

$x \geq -\frac{2}{3}$



$\therefore$ Solution set is $\left\{x: -\frac{2}{3} \leq x \leq \frac{10}{3}\right\}$.

Alternative Method:

$|3x - 4| \leq 6$

Use the algebraic property: $|x|^2 = x^2$.

Squaring both sides:

$$(3x - 4)^2 \leq (6)^2$$

$$9x^2 - 24x + 16 \leq 36$$

$$9x^2 - 24x + 16 - 36 \leq 0$$

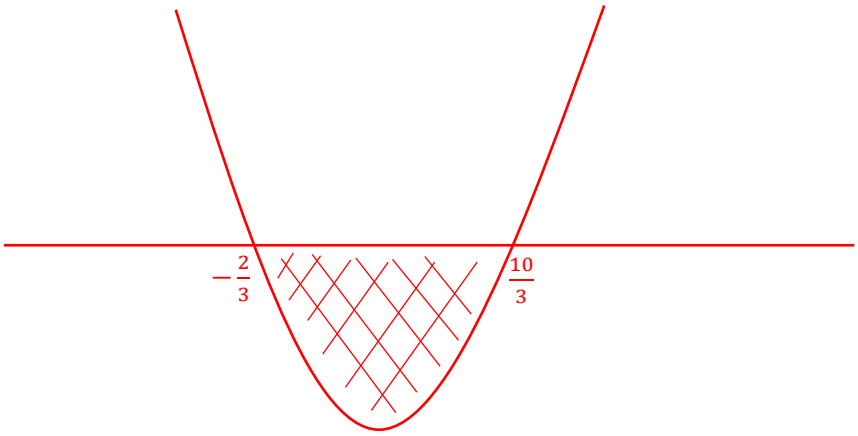
$$9x^2 - 24x - 20 \leq 0$$

$$9x^2 - 30x + 6x - 20 \leq 0$$

$$3x(3x - 10) + 2(3x - 10) \leq 0$$

$$(3x + 2)(3x - 10) \leq 0$$

The critical values are $-\frac{2}{3}$ and $\frac{10}{3}$.



We are looking at the area where $(3x+2)(3x-10) \leq 0$.

$\therefore$ Solution set is $\left\{x: -\frac{2}{3} \leq x \leq \frac{10}{3}\right\}$.

(c) A binary operation is defined on the set of rational numbers by $a * b = \frac{ab}{2}$.

Prove that $*$ is commutative. [3]

$a * b = \frac{ab}{2}$

$= \frac{ba}{2}$

$= b * a$

Since $a * b = b * a$, then $*$ is commutative.

Q.E.D.

(d) Use mathematical induction to prove that $5^n - 1$ is divisible by 4 for $n \in \mathbb{N}$. [8]

Let $P(n)$ be the statement:

$5^n - 1$ is divisible by 4 for $n \in \mathbb{N}$

Looking at $P(1)$.

$5^1 - 1 = 4$ which is divisible by 4.

$\therefore P(n)$ is true for $n = 1$.

Assume $P(k)$ is true.

That is, $5^k - 1$ is divisible by 4

$\therefore 5^k - 1 = 4m$, where $m \in \mathbb{N}$

$\Rightarrow 5^k = 4m + 1$

Looking at $P(k+1)$.

That is, $5^{k+1} - 1$ is divisible by 4.

We need to prove $P(k+1)$.

$$5^{k+1} - 1 = 5^k \cdot 5 - 1$$

$$= (4m+1)(5) - 1$$

$$= 20m + 5 - 1$$

$$= 20m + 4$$

$$= 4(5m+1) \quad \text{which is divisible by 4}$$

$\therefore$ If $P(k)$ is true, $P(k+1)$ is also true.

Since $P(1)$ is true, and since $P(k+1)$ is true when $P(k)$ is true, then by the principle of mathematical induction, $P(n)$ is true for all n, ie.

$5^n - 1$ is divisible by 4 for all $n \in \mathbb{N}$.

Q.E.D.

Total: 25 marks

June 2019 – Question 2 – Solution

(a) Let $A = \{x : x \in \mathbb{R}, x \geq 1\}$

A function $f : A \to \mathbb{R}$ is defined as $f(x) = x^2 - x$. Show that f is one-to-one. [7]

$f(x) = x^2 - x$

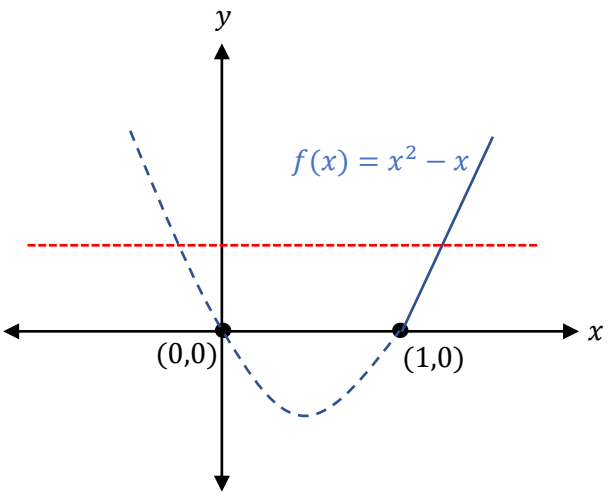
Graphical Proof:

When $f(x) = 0$,

$x^2 - x = 0$

$x(x-1) = 0$

So, the roots are $x = 0$ and $x = 1$.



Any line drawn parallel to the x-axis cuts the graph at most once.

Hence, f is one-to-one.

Q.E.D.

Proof by definition:

$f(x) = x^2 - x$

Now,

$f(a) = a^2 - a$

$f(b) = b^2 - b$

Suppose that $f(a) = f(b)$.

Then, we have

$$a^2 - a = b^2 - b$$

$$a^2 - b^2 - a + b = 0$$

$$(a - b)(a + b) - (a - b) = 0$$

$$(a - b)(a + b - 1) = 0$$

Either $a - b = 0$ or $a + b - 1 = 0$

$a = b$ $a = 1 - b$ which is invalid since $x \geq 1$

Since $f(a) = f(b)$ implies that $a = b$, then f is one-to-one.

Q.E.D.

(b) Let $f(x) = 3x + 1$ and $g(x) = e^{3x}$.

(i) Determine f^{-1}. [2]

$f(x) = 3x + 1$

Let $y = f(x)$.

$y = 3x + 1$

Interchanging the variables x and y.

$x = 3y + 1$

Making y the subject of the formula.

$3y = x - 1$

$y = \frac{x-1}{3}$

$\therefore f^{-1}(x) = \frac{x-1}{3}$

(ii) Determine $f^{-1} \circ g$. [2]

$f^{-1}(x) = \frac{x-1}{3}$, $g(x) = e^{3x}$

Now,

$f^{-1} \circ g = f^{-1}[g(x)]$

$= \frac{e^{3x}-1}{3}$

$= \frac{1}{3}(e^{3x} - 1)$

(c) Solve the equation $3 - \frac{4}{9^x} - \frac{4}{81^x} = 0$. [7]

$$3 - \frac{4}{9^x} - \frac{4}{81^x} = 0$$

$$3 - \frac{4}{(9^x)} - \frac{4}{(9^x)^2} = 0$$

Let $u = 9^x$. Then,

$$3 - \frac{4}{u} - \frac{4}{u^2} = 0$$

$(\times u^2)$

$$3u^2 - 4u - 4 = 0$$

$$(3u + 2)(u - 2) = 0$$

Either $3u + 2 = 0$ or $u - 2 = 0$

$u = -\frac{2}{3}$ $u = 2$

When $u = -\frac{2}{3}$,

$$9^x = -\frac{2}{3}$$

Taking logs on both sides:

$$\log 9^x = \log\left(-\frac{2}{3}\right)$$

$\Rightarrow$ No real solutions exist for x

When $u = 2$,

$9^x = 2$

Taking logs on both sides:

$\log 9^x = \log 2$

$x \log 9 = \log 2$

$x = \frac{\log 2}{\log 9}$

$\therefore\ x = \frac{\log 2}{\log 9}$

(d) Let the roots of the equation $2x^3 - 5x^2 + 4x + 6 = 0$ be α, β and γ.

(i) State the values of $\alpha + \beta + \gamma$, $\alpha\beta + \alpha\gamma + \beta\gamma$ and $\alpha\beta\gamma$. [3]

$2x^3 - 5x^2 + 4x + 6 = 0$

Dividing throughout by 2 gives:

$x^3 - \frac{5}{2}x^2 + 2x + 3 = 0$

which is of the form

$x^3 - (sum\ of\ roots)x^2 + (sum\ of\ the\ product\ of\ two\ of\ the\ roots\ at\ a\ time)x - (product\ of\ the\ roots) = 0$

Therefore,

$\alpha + \beta + \gamma = \frac{5}{2}$

$\alpha\beta + \beta\gamma + \alpha\gamma = 2$

$\alpha\beta\gamma = -3$

(ii) Hence, or otherwise, determine the equation with roots α^2, β^2 and γ^2. [4]

Note: $(\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

For new roots α^2, β^2 and γ^2:

Sum of the roots:

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= \left(\frac{5}{2}\right)^2 - 2(2)$$

$$= \frac{25}{4} - 4$$

$$= \frac{9}{4}$$

Sum of the product of two roots at a time:

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2 = (\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2$$

$$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$= (2)^2 - 2(-3)\left(\frac{5}{2}\right)$$

$$= 4 + 15$$

$$= 19$$

Product of the roots:

$$\alpha^2\beta^2\gamma^2 = (\alpha\beta\gamma)^2$$

$$= (-3)^2$$

$$= 9$$

$\therefore$ The new equation is:

$x^3 - \frac{9}{4}x^2 + 19x - 9 = 0$

$(\times 4)$

$4x^3 - 9x^2 + 76x - 36 = 0$

Total: 25 marks

June 2019 – Question 3 – Solution

(a) Solve the equation $5\sec\theta - 2\sec^2\theta = \tan^2\theta - 1$ for $-\pi \le \theta \le \pi$. [7]

Well,

$$5\sec\theta - 2\sec^2\theta = \tan^2\theta - 1$$

$$2\sec^2\theta - 5\sec\theta + \tan^2\theta - 1 = 0$$

$$2\sec^2\theta - 5\sec\theta + \sec^2\theta - 1 - 1 = 0$$

$$3\sec^2\theta - 5\sec\theta - 2 = 0$$

$$(3\sec\theta + 1)(\sec\theta - 2) = 0$$

Either $3\sec\theta + 1 = 0$ or $\sec\theta - 2 = 0$

$\sec\theta = -\frac{1}{3}$ $\sec\theta = 2$

When $\sec\theta = -\frac{1}{3}$,

$$\frac{1}{\cos\theta} = -\frac{1}{3}$$

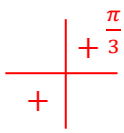
$\cos\theta = -3$ [No solutions since $-1 \le \cos\theta \le 1$]

When $\sec\theta = 2$,

$$\frac{1}{\cos\theta} = 2$$

$$\cos\theta = \frac{1}{2}$$

$$\theta = -\frac{\pi}{3}, \frac{\pi}{3}$$



$\therefore \theta = -\frac{\pi}{3}, \frac{\pi}{3}$ for $-\pi \le \theta \le \pi$.

(b) Let $f(x) = \cos x + 2\sin x$.

(i) Express f in the form $R\sin(x+\alpha)$, where α is an acute angle. [5]

$f(x) = \cos x + 2\sin x$

Let $\cos x + 2\sin x = R\sin(x+\alpha)$

$= R\sin x\cos\alpha + R\cos x\sin\alpha$

$= R\cos x\sin\alpha + R\sin x\cos\alpha$

Equating coefficients gives:

$R\sin\alpha = 1$ $\rightarrow$ Eq. 1

$R\cos\alpha = 2$ $\rightarrow$ Eq. 2

Eq. 1 $\div$ Eq. 2 gives:

$\frac{R\sin\alpha}{R\cos\alpha} = \frac{1}{2}$

$\tan\alpha = \frac{1}{2}$

$\alpha = \tan^{-1}\left(\frac{1}{2}\right)$

$\alpha = 0.464^c$

$R = \sqrt{1^2 + 2^2}$

$= \sqrt{1+4}$

$= \sqrt{5}$

$\therefore \cos x + 2\sin x = \sqrt{5}\sin(x + 0.464^c)$

(ii) Hence, or otherwise, find the general solution of $\cos x + 2\sin x = 0$. [3]

Well,

$$\cos x + 2\sin x = 0$$

$$\sqrt{5}\sin(x + 0.464^c) = 0$$

$$\sin(x + 0.464^c) = 0$$

$$PV = 0$$

G.S. is $x + 0.464^c = n\pi + (-1)^n(0), \quad n \in \mathbb{Z}$

$$x + 0.464^c = n\pi, \qquad n \in \mathbb{Z}$$

$$x = n\pi - 0.464^c, \qquad n \in \mathbb{Z}$$

(iii) Determine the **minimum** value of $\frac{2}{2-f}$. [3]

Well,

$$\frac{2}{2-f} = \frac{2}{2-\sqrt{5}\sin(x+0.464^c)}$$

Now,

$$-1 \leq \sin x \leq 1$$

$$-1 \leq \sin(x + 0.464^c) \leq 1$$

$$\sqrt{5} \geq -\sqrt{5}\sin(x + 0.464^c) \geq -\sqrt{5}$$

$$-\sqrt{5} \leq -\sqrt{5}\sin(x + 0.464^c) \leq \sqrt{5}$$

$$2 - \sqrt{5} \leq 2 - \sqrt{5}\sin(x + 0.464^c) \leq 2 + \sqrt{5}$$

$$\frac{2}{2-\sqrt{5}} \geq \frac{2}{2-\sqrt{5}\sin(x+0.464^c)} \geq \frac{2}{2+\sqrt{5}}$$

$$\frac{2}{2+\sqrt{5}} \leq \frac{2}{2-\sqrt{5}\sin(x+0.464^c)} \leq \frac{2}{2-\sqrt{5}}$$

$\therefore$ The minimum value of $\frac{2}{2-f}$ is $\frac{2}{2+\sqrt{5}}$.

(c) Prove the identity $\tan(A+B) - \tan A = \frac{\sin B}{\cos A \cos(A+B)}$. [7]

Taking L.H.S:

$$\tan(A+B) - \tan A = \frac{\sin(A+B)}{\cos(A+B)} - \frac{\sin A}{\cos A}$$

$$= \frac{\sin A \cos B + \cos A \sin B}{\cos(A+B)} - \frac{\sin A}{\cos A}$$

$$= \frac{\cos A(\sin A \cos B + \cos A \sin B) - \sin A \cos(A+B)}{\cos A \cos(A+B)}$$

$$= \frac{\cos A \sin A \cos B + \cos^2 A \sin B - \sin A(\cos A \cos B - \sin A \sin B)}{\cos A \cos(A+B)}$$

$$= \frac{\cos A \sin A \cos B + \cos^2 A \sin B - \sin A \cos A \cos B + \sin^2 A \sin B}{\cos A \cos(A+B)}$$

$$= \frac{\cos^2 A \sin B + \sin^2 A \sin B}{\cos A \cos(A+B)}$$

$$= \frac{\sin B(\cos^2 A + \sin^2 A)}{\cos A \cos(A+B)}$$

$$= \frac{\sin B(1)}{\cos A \cos(A+B)}$$

$$= \frac{\sin B}{\cos A \cos(A+B)}$$

$$= \text{R.H.S.}$$

$$\therefore \tan(A+B) - \tan A = \frac{\sin B}{\cos A \cos(A+B)}$$

Q.E.D.

Total: 25 marks

June 2019 – Question 4 – Solution

(a) A circle with equation $x^2 + y^2 - 2x - 4y - 5 = 0$ has centre C. A straight line with equation $y = 2x + 5$ intersects the circle at two points, A and B.

(i) Determine the coordinates of C. [3]

$$x^2 + y^2 - 2x - 4y - 5 = 0$$

$$x^2 - 2x + 1 + y^2 - 4y + 4 = 5 + 1 + 4$$

$$(x - 1)^2 + (y - 2)^2 = 10$$

which is of the form $(x - a)^2 + (y - b)^2 = r^2$

where the centre, $C(1, 2)$ and radius, $r = \sqrt{10}$ units.

$\therefore$ The coordinates of C are $(1, 2)$.

(ii) Show that the coordinates of A and B are $(0, 5)$ and $(-2, 1)$ and that BC is perpendicular to AC. [8]

We have,

$(x - 1)^2 + (y - 2)^2 = 10$ $\rightarrow$ Equation 1

$y = 2x + 5$ $\rightarrow$ Equation 2

Substituting Equation 2 into Equation 1 gives:

$$(x-1)^2 + (2x+5-2)^2 = 10$$

$$(x-1)^2 + (2x+3)^2 = 10$$

$$x^2 - 2x + 1 + 4x^2 + 12x + 9 = 10$$

$$5x^2 + 10x = 0$$

$$x^2 + 2x = 0$$

$$x(x+2) = 0$$

Either $x = 0$ or $x + 2 = 0$

$$x = -2$$

When $x = 0$,

$$y = 2(0) + 5$$

$$= 0 + 5$$

$$= 5$$

When $x = -2$,

$$y = 2(-2) + 5$$

$$= -4 + 5$$

$$= 1$$

$\therefore$ The coordinates of A and B are $(0, 5)$ and $(-2, 1)$.

Q.E.D.

Now, in order to show that BC is perpendicular to AC, we show that the product of their gradients is equal to -1.

Points are $B(-2,1)$ and $C(1,2)$.

Gradient of $BC = \frac{y_2-y_1}{x_2-x_1}$

$= \frac{2-1}{1-(-2)}$

$= \frac{1}{3}$

Points are $A(0,5)$ and $C(1,2)$.

Gradient of $AC = \frac{2-5}{1-0}$

$= \frac{-3}{1}$

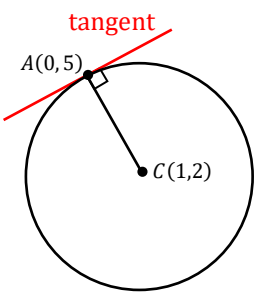
$= -3$

Since $\frac{1}{3} \times -3 = -1$, then BC is perpendicular to AC.

Q.E.D.

(iii) Determine the equation of the tangent to the circle at either Point A or Point B. [2]

Consider the tangent to the circle at Point A.



Gradient of $AC = -3$

Gradient of tangent at Point $A = \frac{-1}{-3}$

$= \frac{1}{3}$

Substituting $m = \frac{1}{3}$ and point $(0, 5)$ into $y - y_1 = m(x - x_1)$ gives:

$y - 5 = \frac{1}{3}(x - 0)$

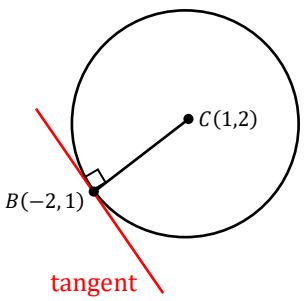
$y - 5 = \frac{1}{3}x$

$y = \frac{1}{3}x + 5$

$3y = x + 15$

$\therefore$ The equation of the tangent to the circle at Point A is: $3y = x + 15$

Consider the tangent to the circle at Point A.



Gradient of $BC = \frac{1}{3}$

Gradient of tangent at Point $B = \frac{-1}{\frac{1}{3}}$

$= -3$

Substituting $m = -3$ and point $(-2, 1)$ into $y - y_1 = m(x - x_1)$ gives:

$y - 1 = -3\big(x - (-2)\big)$

$y - 1 = -3(x + 2)$

$y - 1 = -3x - 6$

$y = -3x - 6 + 1$

$y = -3x - 5$

$\therefore$ The equation of the tangent to the circle at Point B is: $y = -3x - 5$

(b) The points $P(3, -1, 2)$, $Q(1, 2, -4)$ and $R(-1, 1, -2)$ are three vertices of a parallelogram $PQRS$.

(i) Express the vectors $\overrightarrow{PQ}$ and $\overrightarrow{QR}$ in the form $x\hat{\boldsymbol{i}} + y\hat{\boldsymbol{j}} + z\hat{\boldsymbol{k}}$. [3]

Required to express the vectors $\overrightarrow{PQ}$ and $\overrightarrow{QR}$ in the form $x\hat{\boldsymbol{i}} + y\hat{\boldsymbol{j}} + z\hat{\boldsymbol{k}}$.

The points are $P(3,-1,2)$, $Q(1,2,-4)$ and $R(-1,1,-2)$.

So, we have

$\overrightarrow{OP} = 3\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + 2\hat{\boldsymbol{k}}$

$\overrightarrow{OQ} = \hat{\boldsymbol{\imath}} + 2\hat{\boldsymbol{\jmath}} - 4\hat{\boldsymbol{k}}$

$\overrightarrow{OR} = -\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 2\hat{\boldsymbol{k}}$

Now,

Using the Triangle law, we have,

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$

$$= \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} - \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1-3 \\ 2-(-1) \\ -4-2 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ 3 \\ -6 \end{pmatrix}$$

$= -2\hat{\boldsymbol{\imath}} + 3\hat{\boldsymbol{\jmath}} - 6\hat{\boldsymbol{k}}$ which is in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}$

where $x = -2$, $y = 3$ and $z = -6$

Additionally,

$\overrightarrow{QR} = \overrightarrow{OR} - \overrightarrow{OQ}$

$$= \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix}$$

$$= \begin{pmatrix} -1-1 \\ 1-2 \\ -2-(-4) \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$$

$= -2\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + 2\hat{\boldsymbol{k}}$ which is in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}$

where $x = -2$, $y = -1$ and $z = 2$

(ii) Show that the vector $\boldsymbol{s} = -16\hat{\boldsymbol{\jmath}} - 8\hat{\boldsymbol{k}}$ is perpendicular to the plane through P, Q and R. [5]

If $\boldsymbol{s} = -16\hat{\boldsymbol{\jmath}} - 8\hat{\boldsymbol{k}}$ is perpendicular to the plane through P, Q and R, then

$\boldsymbol{s} \cdot \overrightarrow{PQ} = 0$ or $\boldsymbol{s} \cdot \overrightarrow{QR} = 0$.

Well,

$$\boldsymbol{s} \cdot \overrightarrow{PQ} = \begin{pmatrix} 0 \\ -16 \\ -8 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 3 \\ -6 \end{pmatrix}$$

$$= (0)(-2) + (-16)(3) + (-8)(-6)$$

$$= 0 - 48 + 48$$

$$= 0$$

Alternatively,

$$\boldsymbol{s} \cdot \overrightarrow{QR} = \begin{pmatrix} 0 \\ -16 \\ -8 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$$

$$= (0)(-2) + (-16)(-1) + (-8)(2)$$

$$= 0 + 16 - 16$$

$$= 0$$

$\therefore$ The vector $\boldsymbol{s} = -16\hat{\boldsymbol{j}} - 8\hat{\boldsymbol{k}}$ is perpendicular to the plane through P, Q and R.

Q.E.D.

(iii) Hence, determine the Cartesian equation of the plane through P, Q and R. [4]

Required to determine the Cartesian equation of the plane through P, Q and R.

Let $\boldsymbol{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

From the point $P(3, -1, 2)$, let $\boldsymbol{a} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$.

From vector $\boldsymbol{s} = -16\hat{\boldsymbol{j}} - 8\hat{\boldsymbol{k}}$, let the perpendicular vector of the plane be

$\boldsymbol{n} = \begin{pmatrix} 0 \\ -16 \\ -8 \end{pmatrix}$.

Hence, the Cartesian equation of the plane is:

$$\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -16 \\ -8 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -16 \\ -8 \end{pmatrix}$$

$$-16y - 8z = (3)(0) + (-1)(-16) + (2)(-8)$$

$$-16y - 8z = 0 + 16 - 16$$

$$-16y - 8z = 0$$

$$2y + z = 0$$

Total: 25 marks

June 2019 – Question 5 – Solution

(a) A function is given as $f(x) = \begin{cases} \sqrt{-x} & x < 0 \\ 1 & 0 < x \leq 1 \\ \sqrt{x} & x > 1 \end{cases}$

(i) Determine $\lim_{x \to 0^-} f(x)$. [2]

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \sqrt{-x}$$

$$= \sqrt{0}$$

$$= 0$$

(ii) Determine whether $\lim_{x \to 0} f(x)$ exists. **Give a reason for your answer.** [3]

For $\lim_{x \to 0} f(x)$ to exist, $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x)$.

Now,

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} 1$$

$$= 1$$

And $\lim_{x \to 0^-} f(x) = 0$.

Since $\lim_{x \to 0^-} f(x) = 0$ and $\lim_{x \to 0^+} f(x) = 1$, then $\lim_{x \to 0^-} f(x) \neq \lim_{x \to 0^+} f(x)$.

$\therefore$ The $\lim_{x \to 0} f(x)$ does not exist.

(iii) Determine whether f is continuous at $x = 1$. **Give a reason for your answer.** [3]

Well, $f(1) = 1$.

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} 1$$

$$= 1$$

And,

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} \sqrt{x}$$

$$= \sqrt{1}$$

$$= 1$$

Since $\lim_{x \to 1^-} f(x) = 1 = \lim_{x \to 1^+} f(x)$, then $\lim_{x \to 1} f(x) = 1$.

So, $\lim_{x \to 1} f(x)$ exists.

Since $\lim_{x \to 1} f(x) = 1$ and $f(1) = 1$, then $\lim_{x \to 1} f(x) = f(1)$.

All three conditions for continuity are satisfied.

$\therefore f$ is continuous at $x = 1$.

(b) Determine the point at which the tangent to the curve $y = x\sqrt{x}$ is parallel to the line $3x - y + 6 = 0$. [4]

Well,

$3x - y + 6 = 0$

$y = 3x + 6$

Gradient of line = 3.

Now,

$y = x\sqrt{x}$

$y = x(x)^{\frac{1}{2}}$

$y = x^{\frac{3}{2}}$

$\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}}$

Note that parallel lines have the same gradient.

So, we have

$\frac{3}{2}x^{\frac{1}{2}} = 3$

$x^{\frac{1}{2}} = 2$

Squaring both sides gives:

$x = 4$

When $x = 4$,

$y = 4\sqrt{4}$

$= 4(2)$

$= 8$

$\therefore$ The point at which the tangent to the curve is parallel to the line is $(4, 8)$.

(c) A function is given as $y = \sin^2(\cos x)$.

(i) Show that the x-coordinates of the stationary values of y are

- $n\pi, n \in \mathbb{Z}$
- $\pm\frac{\pi}{2} + 2n\pi, n \in \mathbb{Z}$ [9]

$y = \sin^2(\cos x)$

Let $u = \cos x$ $\qquad$ $y = \sin^2 u$

$\frac{du}{dx} = -\sin x$ $\qquad$ $\frac{dy}{du} = 2\cos u \sin u$

Using the chain rule,

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$= 2\cos u \sin u\,(-\sin x)$

$= 2\sin u \cos u\,(-\sin x)$

$= \sin 2u\,(-\sin x)$

$= [\sin(2\cos x)](-\sin x)$

Note that:

$\sin 2x = 2\sin x \cos x$

At stationary points, $\frac{dy}{dx} = 0$.

When $\frac{dy}{dx} = 0$,

$[\sin(2\cos x)](-\sin x) = 0$

Either $\sin(2\cos x) = 0$ or $-\sin x = 0$.

Now,

$-\sin x = 0$

$\sin x = 0$

$PV = 0$

G.S. is $x = n\pi \pm (-1)^n(0) \qquad , n \in \mathbb{Z}$

$x = n\pi \qquad , n \in \mathbb{Z}$

Now,

$\sin(2\cos x) = 0$

$2\cos x = 0$

$\cos x = 0$

$PV = \frac{\pi}{2}$

G.S. is $x = 2n\pi \pm \frac{\pi}{2}$ $\quad , n \in \mathbb{Z}$

$x = \pm\frac{\pi}{2} + 2n\pi$ $\quad , n \in \mathbb{Z}$

$\therefore$ The x-coordinates of the stationary values of y are

- $n\pi, n \in \mathbb{Z}$
- $\pm\frac{\pi}{2} + 2n\pi, n \in \mathbb{Z}$

Q.E.D.

(ii) Determine the nature of the stationary values at $x = n\pi, n \in \mathbb{Z}$. [4]

$\frac{dy}{dx} = [\sin(2\cos x)](-\sin x)$

$\frac{dy}{dx} = -\sin x \sin(2\cos x)$

Let $u = -\sin x$ $\quad$ $v = \sin(2\cos x)$

$\frac{du}{dx} = -\cos x$ $\quad$ $\frac{dv}{dx} = -2\sin x\cos(2\cos x)$

Now,

$\frac{d^2y}{dx^2} = u\frac{dv}{dx} + v\frac{du}{dx}$

$= (-\sin x)[-2\sin x\cos(2\cos x)] + \sin(2\cos x)(-\cos x)$

$= 2\sin^2 x\cos(2\cos x) - \cos x\sin(2\cos x)$

When $x = n\pi$ where n is odd,

$$\frac{d^2y}{dx^2} = 2\sin^2 n\pi \cos(2\cos n\pi) - \cos n\pi \sin(2\cos n\pi)$$

$$= 0 - (-1)\sin\big(2(-1)\big)$$

$$= \sin(-2) < 0$$

Since $\frac{d^2y}{dx^2} < 0$, then there is a maximum stationary point.

When $x = n\pi$ where n is even,

$$\frac{d^2y}{dx^2} = 2\sin^2 n\pi \cos(2\cos n\pi) - \cos n\pi \sin(2\cos n\pi)$$

$$= 0 - (1)\sin\big(2(1)\big)$$

$$= -\sin(2) < 0$$

Since $\frac{d^2y}{dx^2} < 0$, then there is a maximum stationary point.

$\therefore$ The nature of the stationary values at $x = n\pi, n \in \mathbb{Z}$ is maximum.

Total: 25 marks

June 2019 – Question 6 – Solution

(a) By using the substitution $u = 1 - x$, determine $\int x(1-x)^2\, dx$. [5]

Let $u = 1 - x$ Also, $x = 1 - u$.

$$\frac{du}{dx} = -1$$

$$dx = -du$$

Now,

$$\int x(1-x)^2\, dx = \int (1-u)u^2\,(-1)\, du$$

$$= \int (u-1)u^2\, du$$

$$= \int (u^3 - u^2)\, du$$

$$= \frac{u^4}{4} - \frac{u^3}{3} + c$$

$$= \frac{(1-x)^4}{4} - \frac{(1-x)^3}{3} + c$$

(b) Given that $f(t) = 2\cos t$ and $g(t) = 4\sin 5t + 3\cos t$, show that

$$\int [f(t) + g(t)]\, dt = \int f(t)\, dt + \int g(t)\, dt \qquad [5]$$

Taking L.H.S:

$$\int [f(t) + g(t)]\, dt = \int (2\cos t + 4\sin 5t + 3\cos t)\, dt$$

$$= \int (4\sin 5t + 5\cos t)\, dt$$

$$= 4\int \sin 5t\, dt + 5\int \cos t\, dt$$

$$= 4\left(-\frac{1}{5}\right)\cos 5t + 5\sin t + c$$

$$= -\frac{4}{5}\cos 5t + 5\sin t + c$$

Taking R.H.S:

$$\int f(t)\, dt + \int g(t)\, dt = \int (2\cos t)\, dt + \int (4\sin 5t + 3\cos t)\, dt$$

$$= 2\int \cos t\, dt + 4\int \sin 5t\, dt + 3\int \cos t\, dt$$

$$= 2\sin t + 4\left(-\frac{1}{5}\right)\cos 5t + 3\sin t + c$$

$$= -\frac{4}{5}\cos 5t + 5\sin t + c$$

$$= \text{L.H.S.}$$

$$\therefore \int [f(t) + g(t)]\, dt = \int f(t)\, dt + \int g(t)\, dt$$

Q.E.D.

(c) Calculate the volume of the solid formed by rotating the region enclosed by $y = \sin x$, $x = \frac{\pi}{2}$, $x = \pi$ and $y = 0$ about the x-axis. [6]

Volume about the x-axis $= \pi \int_{\frac{\pi}{2}}^{\pi} y^2 \, dx$

Now,

$y = \sin x$

$y^2 = (\sin x)^2$

$y^2 = \sin^2 x$

Recall that,

$\cos 2x = 1 - 2\sin^2 x$

$2\sin^2 x = 1 - \cos 2x$

$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

So, we have,

$V = \pi \int_{\frac{\pi}{2}}^{\pi} y^2 \, dx$

$= \pi \int_{\frac{\pi}{2}}^{\pi} \sin^2 x \, dx$

$= \pi \int_{\frac{\pi}{2}}^{\pi} \frac{1}{2}(1 - \cos 2x) \, dx$

$$= \frac{\pi}{2}\int_{\frac{\pi}{2}}^{\pi}(1-\cos 2x)\,dx$$

$$= \frac{\pi}{2}\left[x - \frac{1}{2}\sin 2x\right]_{\frac{\pi}{2}}^{\pi}$$

$$= \frac{\pi}{2}\left[\left(\pi - \frac{1}{2}\sin 2\pi\right) - \left(\frac{\pi}{2} - \frac{1}{2}\sin 2\left(\frac{\pi}{2}\right)\right)\right]$$

$$= \frac{\pi}{2}\left[\pi - \frac{1}{2}(0) - \left(\frac{\pi}{2} - \frac{1}{2}\sin \pi\right)\right]$$

$$= \frac{\pi}{2}\left[\pi - \frac{1}{2}(0) - \frac{\pi}{2} + \frac{1}{2}(0)\right]$$

$$= \frac{\pi}{2}\left(\pi - \frac{\pi}{2}\right)$$

$$= \frac{\pi}{2}\left(\frac{\pi}{2}\right)$$

$$= \frac{\pi^2}{4}\text{ units}^3$$

(d) Solve the differential equation $\frac{dy}{dx} = \frac{x\sqrt{x^2+1}}{y}$, given that when $x = 0$, $y = 2$. [9]

$$\frac{dy}{dx} = \frac{x\sqrt{x^2+1}}{y}$$

$$y\,dy = x\sqrt{x^2+1}\,dx$$

Integrating both sides:

$$\int y\,dy = \int x\sqrt{x^2+1}\,dx$$

Consider $\int x\sqrt{x^2+1}\,dx$.

Let $u = x^2 + 1$

$$\frac{du}{dx} = 2x$$

$$dx = \frac{du}{2x}$$

Now,

$$\int x\sqrt{x^2+1}\,dx = \int x\sqrt{u}\frac{du}{2x}$$

$$= \frac{1}{2}\int u^{\frac{1}{2}}\,du$$

$$= \frac{\frac{1}{2}u^{\frac{3}{2}}}{\frac{3}{2}} + c$$

$$= \frac{1}{3}u^{\frac{3}{2}} + c$$

$$= \frac{1}{3}(x^2+1)^{\frac{3}{2}} + c$$

So, we have

$$\int y\,dy = \int x\sqrt{x^2+1}\,dx$$

$$\frac{y^2}{2} = \frac{1}{3}(x^2+1)^{\frac{3}{2}} + c$$

When $x = 0$ and $y = 2$,

$$\frac{(2)^2}{2} = \frac{1}{3}[(0)^2 + 1]^{\frac{3}{2}} + c$$

$$2 = \frac{1}{3}(1)^{\frac{3}{2}} + c$$

$$2 = \frac{1}{3} + c$$

$$c = 2 - \frac{1}{3}$$

$$c = \frac{5}{3}$$

Hence,

$$\frac{y^2}{2} = \frac{1}{3}(x^2 + 1)^{\frac{3}{2}} + \frac{5}{3}$$

$$y^2 = \frac{2}{3}(x^2 + 1)^{\frac{3}{2}} + \frac{10}{3}$$

$\therefore$ The equation is: $y^2 = \frac{2}{3}(x^2 + 1)^{\frac{3}{2}} + \frac{10}{3}$

Total: 25 marks